Cloudflare R2 Synchronization Architecture

Implementation & Script Progression Report

1. Environment Setup & Package Installation

To prepare the system for the synchronization scripts, the Python virtual environment requires the following core dependencies to be installed via the terminal:

pip install watchdog: Provides real-time monitoring of the local Windows file system to capture user interactions instantly.

pip install boto3: The standard AWS SDK for Python, utilized to communicate directly with the S3 compatible Cloudflare R2 API.

pip install python-dotenv: Allows the script to securely load environmental variables and secret keys, keeping credentials out of the source code.

2. System Configuration & Architecture

The system connects the local Python environment to the cloud bucket using a secure .env file containing the following architecture parameters:

Credentials: **R2_ACCOUNT_ID**, **R2_ACCESS_KEY_ID**, **R2_SECRET_ACCESS_KEY**

Local Target: **WATCH_DIR** is explicitly set to **C:\data\incoming**.

Cloud Target: **R2_BUCKET** is set to **filesync**, utilizing the **R2_PREFIX** of **incoming**.

The Python scripts initialize this connection by explicitly directing the **boto3.client** endpoint URL to **https://{R2_ACCOUNT_ID}r2.cloudflarestorage.com** instead of the default AWS servers.

3. Execution & Workflow

Execution Command: Each script is initialized via the terminal using standard Python execution
Ex: python 5sync_bi_all.py

Background Process: Once running, the script operates continuously in the background until manually terminated (Ctrl+C).

File Name	Sync Direction	Action	Action Description
1sync.py	**Local → R2**	**Add**	Detects new files dropped specifically into the root local folder and pushes them to R2.
2sync_sf.py	**Local → R2**	**Add**	Extends the initial sync with recursive capabilities to monitor and upload nested subdirectories (e.g., home/new_folder).
3sync_all.py	**Local → R2**	**Add**	Uploads newly created local files and folders to the cloud bucket.
		Edit	Detects saved modifications to local files and uploads the new version, overwriting the older R2 object.
		Rename	Uploads the file under its new name to R2, and explicitly deletes the old object key to prevent duplicates.
		Location Change	Uploads the file to the new R2 folder path and deletes the old cloud object when moved locally.
		Delete	Acts as a true one-way mirror by detecting local file removals and deleting the corresponding object from R2.
4sync_bi.py	**Local → R2**	**Add**	Uploads newly created local files to the cloud bucket using Watchdog.
		Edit	Uploads modified local files to R2, overwriting the older object. (One-way only).
		Rename	Uploads local renames to the new R2 path and deletes the old cloud object. (One-way only).
		Location Change	Uploads local moves to the new R2 path and deletes the old cloud object. (One-way only).
		Delete	Mirrors local deletions by deleting the corresponding R2 object. (One-way only).
	R2 → Local	**Add**	A background polling thread fetches the R2 file list and downloads new/missing files down to the local PC.

File Name	Sync Direction	Action	Action Description
5sync_bi_all.py	**Local → R2**	**Add**	Syncs new files created locally to the R2 bucket.
		Edit	Uploads modified local files to R2, completely overwriting the older cloud object.
		Rename	Uploads the renamed local file to R2 and deletes the old cloud object key.
		Location Change	Uploads the moved local file to its new R2 directory path and deletes the old cloud object.
		Delete	Removes the corresponding cloud object when a file is deleted from the local PC.
	R2 → Local	**Add**	Polling thread detects new files dropped directly into R2 and downloads them to the PC.
		Edit	Compares timestamps/sizes. If the R2 file is newer/different, it downloads and overwrites the local PC file.
		Rename	Handles R2's native copy-then-delete rename behavior by downloading the newly named file and deleting the old local duplicate.
		Location Change	Downloads the file from the new cloud path and deletes the local file residing at the old path.
		Delete	True mirror deletions: cross-references the local drive and removes any local files that are missing from the R2 bucket.
check_bucket.py	*N/A (Read-Only)*	**List**	Independent auditing tool that queries the R2 API directly to list live bucket contents, confirming successful syncs.